SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	AIML/2024	**Date:**	16-10-2026

Code of Conduct

I, K.Mounith(192425090)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Mounith

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.
Ans: Full-duplex = 2 × 25 kHz = 50 kHz; 45 MHz = 45,000 kHz; 45,000 ÷ 50 = 900 channels

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.
Ans: Channels per cell = Total ÷ N = 560 ÷ 7 = 80 channels

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.
Ans: Total = Carriers × Slots = 100 × 8 = 800 voice channels

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).
Ans: Ratio = 80 ÷ 4 = 20; dB = 10·log10(20) = 13.01 dB

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.
Ans: C/I = C − I = −60 − (−92) = 32 dB

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?
Ans: Total = 42,000 kHz ÷ 300 kHz = 140; per cell = 140 ÷ 7 = 20 channels

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?
Ans: $Pr \propto 1/d^4$; doubling distance → ÷ 2^4 = ÷ 16; 64 ÷ 16 = 4 W

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?
Ans: $Pr \propto 1/d^2$; radius × 2 → loss × 2^2 = 4; transmit power must be 4 times higher (+6 dB)

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?
Ans: Total = 4 cells × 15 channels = 60 channels

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.
Ans: v = 100 ÷ 3.6 = 27.78 m/s; $\lambda = c/f = 3\times10^8 \div 9\times10^8 = 0.333$ m; $fd = v/\lambda = 83.3$ Hz

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?
Ans: Slot = Frame ÷ Slots = 5 ms ÷ 8 = 0.625 ms

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?
Ans: Carriers = 24,000 kHz ÷ 200 kHz = 120 carriers

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.
Ans: Pt = 10·log10(8000 mW) = 39.03 dBm; Pr = 39.03 − 33 = 6.03 dBm

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.
Ans: $\lambda = 3\times10^8 \div 1.9\times10^9 = 0.158$ m; PL(1 m) = 20·log(4π/λ) = 38.0 dB; + 20·log(2500) = 67.96; total ≈ 106 dB

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?
Ans: Capacity = Cells × Calls = 6 × 250 = 1500 calls

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?
Ans: Active calls = Rate × Holding time = 4000 × 180 s = 720,000 calls

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?
Ans: Bandwidth = 150 × 25 kHz = 3750 kHz = 3.75 MHz

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?
Ans: Wasted = 30 guard bands × 40 kHz = 1200 kHz = 1.2 MHz

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?
Ans: Distance = 90 km; cell diameter = 2 × 1.5 = 3 km; handoffs = 90 ÷ 3 = 30

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?
Ans: BS = 8 × 40 = 320; users = 320 × 250 = 80,000 users

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?
Ans: Cell area = $R^2 = 0.4^2 = 0.16$ km²; cells = 400 ÷ 0.16 = 2500 cells

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?
Ans: Channels = 6 MHz ÷ 30 kHz = 6000 ÷ 30 = 200 channels

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?
Ans: Calls/h = 1200 × 4 = 4800; minutes = 4800 × 2.5 = 12,000; ÷ 60 = 200 Erlangs

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?
Ans: Users = 40 MHz ÷ 400 kHz = 40,000 ÷ 400 = 100 users

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?
Ans: Capacity = 600 × 1200 = 720,000 users

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?
Ans: Rounds (worst case) = Cells ÷ Cells per round = 200 ÷ 4 = 50 rounds

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?
Ans: Offered = 40,000 × 3 = 120,000; excess = 120,000 − 100,000 = 20,000; blocked = 20,000 ÷ 120,000 ≈ 16.7 %

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?
Ans: Roaming = 15 % × 600,000 = 0.15 × 600,000 = 90,000 subscribers

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?
Ans: Utilization = 9000 ÷ 12,000 = 0.75 = 75 %

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?
Ans: Total = 48,000 ÷ 200 = 240 channels; per cell = 240 ÷ 4 = 60 channels

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?
Ans: Channels = 30 MHz ÷ 200 kHz = 30,000 ÷ 200 = 150 channels

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?
Ans: Successful = 2500 × 0.99 = 2475 calls

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?
Ans: Pt = 10·log10(12,000 mW) = 40.79 dBm; Pr = 40.79 − 45 = −4.21 dBm

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?
Ans: Capacity = 7 cells × 900 = 6300 users

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?
Ans: Slot = 4.8 ms ÷ 8 = 0.6 ms

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?
Ans: $v = 150 \div 3.6 = 41.67$ m/s; $\lambda = 3\times10^8 \div 1.8\times10^9 = 0.1667$ m; $f_d = 41.67 \div 0.1667 = 250$ Hz

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?
Ans: Offered = 70,000 × 4 = 280,000; excess = 280,000 − 250,000 = 30,000; blocked = 30,000 ÷ 280,000 ≈ 10.7 %

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?
Ans: Capacity = 9 cells × 600 = 5400 users

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?
Ans: Dropped = 60,000 × 0.4 % = 60,000 × 0.004 = 240 calls/s

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?
Ans: Path loss = Pt − Pr = 43 − (−87) = 130 dB

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?
Ans: Total area = 150 × 2.5 km² = 375 km²

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?
Ans: Total channels = 48,000 ÷ 200 = 240; per cell = 240 ÷ 6 = 40 channels

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?
Ans: Cycles = Total users ÷ Users per cycle = 150,000 ÷ 6000 = 25 cycles

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?
Ans: Handoffs = Call duration ÷ Interval = 9 ÷ 3 = 3 handoffs

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?
Ans: SNR = S − N = −85 − (−115) = 30 dB

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?
Ans: Bandwidth = 250 × 150 kbps = 37,500 kbps = 37.5 Mbps

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?
Ans: Calls/h = 1500 × 4 = 6000; minutes = 6000 × 2.5 = 15,000; ÷ 60 = 250 Erlangs

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?
Ans: Bandwidth = 250 × 12 kHz = 3000 kHz = 3 MHz

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?
Ans: Throughput = 1500 × 600 kbps = 900,000 kbps = 900 Mbps

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Ans: Area per tower = $\pi R^2 = \pi \times 4^2 = 50.27$ km²; towers = 350 ÷ 50.27 = 6.96 ≈ 7 towers